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displacement of the particle is taken towards left of equilibrium position (x takes negative value), the force (or acceleration) will point towards equilibrium (towards right). This type of force is known as ***restoring force*** *because it always directs the particle executing simple harmonic motion to restore to its original (equilibrium or mean) position.* This force (restoring force) is central and attractive whose center of attraction is the equilibrium position.

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In order to represent in two or three dimensions, we can write using vector notation

$$\vec{F} = -k\vec{r} \qquad (10.4)$$

where $\vec{r}$ is the displacement of the particle from the chosen origin. Note that the force and displacement have a linear relationship. This means that the exponent of force $\vec{F}$ and the exponent of displacement $\vec{r}$ are unity. The sketch between cause (magnitude of force $|\vec{F}|$) and effect (magnitude of displacement $|\vec{r}|$) is a straight line passing through second and fourth quadrant as shown in Figure 10.5. By measuring slope $\frac{1}{k}$, one can find the numerical value of force constant k.

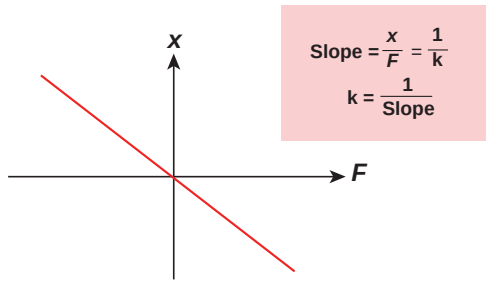


Figure 10.5 Force verses displacement graph

10.2.1 The projection of uniform circular motion on a diameter of SHM

Consider a particle of mass m moving with uniform speed v along the circumference of a circle whose radius is r in anti-clockwise direction (as shown in Figure 10.6). Let us assume that the origin of the coordinate system coincides with the center O of the circle. If ω is the angular velocity of the particle and θ the angular displacement of the particle at any instant of time t, then $\theta = \omega t$. By projecting the uniform circular motion on its diameter gives a simple harmonic motion. This means that we can associate

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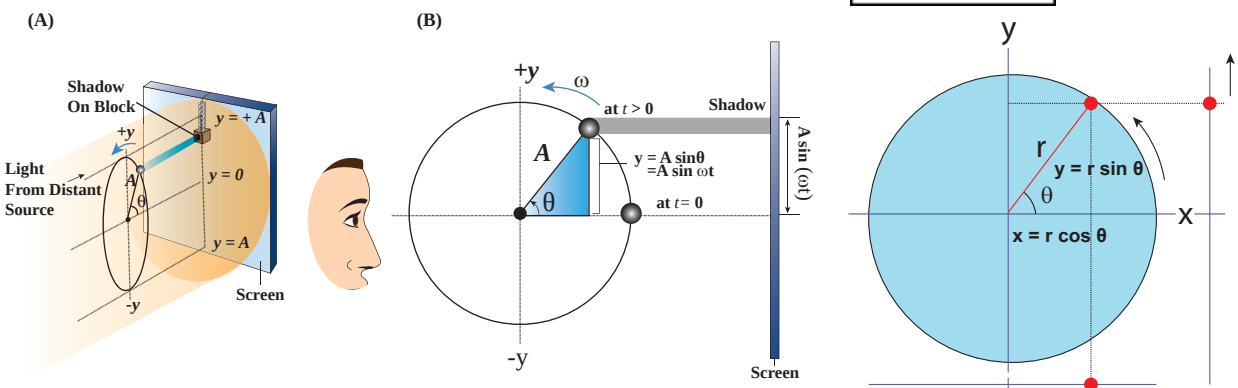


Figure 10.6 Projection of moving particle on a circle on a diameter